

Department of Computer Engineering & Applications							
GLA University, Mathura							
Training cum Learning Delivery Plan							
JOVAC Title:		DevOps					
Name of Faculty Co-ordinators:		Rahul					
Expected No of Lectures:		80-100 Hrs.					
Name of External Agency		Technical Guftgu					
Traing Module (10 June to 31 July)							
Lect	Module	Topic	Pre Reading Material	Sub Topic	Methodology	Learning Outcomes (Lecture Wise)	Prepare Placement oriented Questions(6-8) and Quiz (30 each) on assigned topic
1	Module 1	Linux		What is Linux, RHEL Installation, FHS: File System Hierarchy Standard, Vi & Vim editor, how to create file: touch command, how to create directory: mkdir command, how to paste & cut file & directory: cp & mv command, how to remove file & directory: rm command	PPT + LABS	Gain foundational knowledge of Linux operating systems, commands, and system administration.	Explain the key concepts of Linux, including the installation of Red Hat Enterprise Linux (RHEL), the File System Hierarchy Standard (FHS), and basic file management. How do you use the vi and vim editors to modify files, and how can you manage files and directories using commands like touch , mkdir , cp , mv , and rm ? Provide examples of creating, copying, moving, and deleting files and directories in Linux.
2			youtube.com/watch?v=zA3ymx0G	Permission for File & Directory, Head & Tail command, grep command, find, sed, awk command, User Management & Group Management, Archiving and Transferring Files, Networking	PPT + LABS		Discuss file and directory permissions in Linux and demonstrate how to manage them. How do the head and tail commands work, and what is their use for viewing file contents? Explain the functionality of the grep , find , sed , and awk commands with examples. Additionally, describe the basics of user and group management in Linux. How can files be archived and transferred in Linux? Finally, explain the fundamentals of networking in Linux and how these tools and commands play a role in system administration.
				SSH server, scp & rsync command, Controlling service daemon, Repository server: RPM, YUM, DNF, Storage: Partition, LVM, SWAP, RAID	PPT + LABS		Explain the setup and configuration of an SSH server in Linux. How do the scp and rsync commands differ in transferring files between systems, and when would you use each? Describe the process of controlling service daemons in Linux. What are the roles of RPM , YUM , and DNF in managing software packages on a repository server? Discuss storage management in Linux, including concepts like partitioning , LVM , SWAP , and RAID , and explain their use cases and differences.
3				Webserver: Apache server, Nginx server, File server: NFS, Samba, Database: MariaDB, Security: firewallld, SELinux	PPT + LABS		Compare and contrast the functionalities of Apache and Nginx web servers in terms of performance, use cases, and configuration. How do NFS and Samba differ in providing file-sharing services in Linux? Explain how to set up and manage a MariaDB database in Linux. Additionally, discuss the role of firewalld and SELinux in securing a Linux system, including their configurations and how they help protect the system from unauthorized access and vulnerabilities.
4	Module 2	Shell Scripting	Shell Scripting Tutorial for Beginners	Introduction to Shell Scripting, Types of Shells, Basics of Shell Commands	PPT + LABS	Learn to automate tasks and improve system administration using shell scripts	What is Shell Scripting , and how does it differ from other scripting languages? Discuss the various types of shells available in Linux and their unique features. Additionally, explain the basic shell commands used in scripting, and provide examples of how they are utilized for automating tasks in a Linux environment.
5			Shell Scripting Tutorial for Beginners	Variables and Data Types in Shell Scripting, Input and Output in Shell Scripts	PPT + LABS		How are variables and data types defined and used in shell scripting ? Explain the different types of variables available and how to handle input and output in shell scripts. Provide examples of how to accept user input and display output in a Linux shell environment.
6			Shell Scripting Tutorial for Beginners	Conditional Statements (if-else, switch), Loops (for, while, until)	PPT + LABS		How do conditional statements (such as if-else and switch) work in shell scripting, and when would you use each type? Additionally, explain the different types of loops in shell scripting (for , while , until) and provide examples of how they can be used to automate repetitive tasks in a Linux environment.

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7				Functions in Shell Scripting, File Handling, Error Handling and Debugging	PPT + LABS		How are functions defined and used in shell scripting, and what are their advantages for code reuse and modularity? Explain the process of file handling in shell scripts, including reading, writing, and manipulating files. Additionally, discuss error handling and debugging techniques in shell scripting, and provide examples of how to manage errors and debug scripts effectively in a Linux environment. This question combines all of the key topics into one detailed quer
8	Module 3	Ansible Automation		Introduction about Automation, Understanding Ansible architecture, Ansible Modules and Inventory	PPT + LABS	Understand how to automate IT infrastructure provisioning and configuration management using Ansible.	What is automation in the context of IT operations, and how does it improve efficiency and consistency? Explain the architecture of Ansible , including its key components and how they interact. Additionally, discuss the role of Ansible modules and the inventory system in automating tasks and managing configurations across multiple systems.
9			Automation Course D0407 Class #1	Task Management with Ad-hoc commands, How to write Playbooks, Notify & Handlers in Playbook	PPT + LABS		How does task management work in Ansible using ad-hoc commands , and when should you use them? Explain the process of writing Ansible playbooks to automate tasks, including the structure and syntax. Additionally, describe the use of notify and handlers in playbooks to trigger actions based on specific conditions or events during playbook execution.
10				Roles Structure and Ansible Galaxy, Vault Encryption in Ansible for security	PPT + LABS		What is the role structure in Ansible, and how does it help in organizing and reusing automation tasks? Explain the concept of Ansible Galaxy and how it provides pre-built roles for common tasks. Additionally, discuss the importance of Vault encryption in Ansible and how it ensures the security of sensitive data, such as passwords and keys, within playbooks.
11			Automation Course D0407 Class #1	AWS instance create using Ansible, AWS instance start & stop using Ansible, AWS instance terminate using Ansible	PPT + LABS		How can you use Ansible to create, start, stop , and terminate AWS instances? Explain the steps involved in automating the management of AWS EC2 instances using Ansible, including the necessary modules and playbooks. Provide examples of how these tasks can be accomplished with Ansible to streamline infrastructure management.
12	Module 4	Docker	Docker tutorial in Hindi #1 what	Introduction to Containers, Basics of Containers, Containers Networking	PPT + LABS	Master containerization concepts and the creation, management, and deployment of Docker containers.	What is the concept of containers in software development, and how do they differ from traditional virtualization? Explain the basics of containers , including their structure and use cases. Additionally, discuss container networking and how containers communicate with each other and the external network in a containerized environment.
13			Docker tutorial in Hindi #2 Install	Build Docker Image, Dockerfile	PPT + LABS		How do you build a Docker image , and what is the role of a Dockerfile in this process? Explain the key instructions used in a Dockerfile to define the environment, dependencies, and configurations required for creating a custom Docker image.
14			Docker tutorial in Hindi #3 Install	Docker Storage, Multi-container Applications	PPT + LABS		What is Docker storage , and how does Docker manage data persistence in containers? Additionally, explain the concept of multi-container applications in Docker, and how tools like Docker Compose help in managing and orchestrating containers that work together as part of a single application.
15			Docker tutorial in Hindi #4 What	Docker Registry, Docker Compose	PPT + LABS		What is a Docker Registry , and how does it facilitate the storage and distribution of Docker images? Additionally, explain the role of Docker Compose in managing multi-container applications, and how it simplifies the process of defining and running containers using a single configuration file.
16			What is Kubernetes INTRODUCTION	Introduction to Kubernetes, Kubernetes Installation, Application Deployment	PPT + LABS		What is Kubernetes , and how does it simplify the management of containerized applications at scale? Explain the process of Kubernetes installation , including the necessary steps and tools. Additionally, describe how application deployment works in Kubernetes, including the key components involved and the steps for deploying and managing applications within a Kubernetes cluster.

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17	Module 5	Kubernetes	Kubernetes Architecture in Hindi	Storage Management, Pod Scheduling	PPT + LABS	Learn to deploy, manage, and scale containerized applications using Kubernetes orchestration.	How does storage management work in Kubernetes, and what are the different types of storage resources available for use within a cluster? Additionally, explain the concept of pod scheduling in Kubernetes, including how the scheduler decides where to place pods based on available resources, constraints, and policies.
18			Kubernetes in DevOps in Hindi 	Resource Allocation, Networking	PPT + LABS		How does resource allocation work in Kubernetes, and how are CPU and memory managed for pods? Additionally, explain the basics of Kubernetes networking , including pod communication, services, and network policies.
19			How to create Single Node Cluste	Monitoring Kubernetes, Security	PPT + LABS		How is monitoring implemented in Kubernetes to track the health and performance of clusters and applications? Additionally, explain the key aspects of Kubernetes security , including pod security policies, role-based access control (RBAC), and other mechanisms to protect resources within the cluster.
20	Module 6	AWS Cloud	What is cloud and how it works in	What are AWS Prerequisites and Strategy of Success, AWS Certification Tracks, What is Cloud Computing, Cloud Computing Models (IaaS, PaaS, SaaS), Types of Clouds (Private, Public, Hybrid), Understanding AWS Global Infrastructure: Regions, Availability Zones, AWS Edge Locations and Edge Cache, How to select the best Region	PPT + LABS	Develop proficiency in cloud services and architecture design using AWS.	What are the prerequisites and strategy for success in pursuing AWS certification? Explain the different AWS certification tracks available. Additionally, define cloud computing and describe the various cloud computing models (IaaS, PaaS, SaaS). Discuss the types of clouds (Private, Public, Hybrid), and explain AWS global infrastructure , including regions , availability zones , edge locations , and edge cache . How should you select the best AWS region for your workloads?
21			IAAS, PAAS and SAAS in AWS	Sign-Up for AWS Services, Visit to AWS Control Panel, AWS Free Services and Terminologies, AWS Free Tier, Launch your first EC2 instance (Windows), Launch your first EC2 instance (Linux), AWS Free Tier Resources after Sign-up for 12 months	PPT + LABS		How do you sign up for AWS services and navigate the AWS Control Panel? Describe key AWS services and terminologies, explain the benefits of the AWS Free Tier (including resources available for 12 months), and demonstrate how to launch your first EC2 instance on both Windows and Linux.
22			Types of EC2 Instances Part 1-Hi	VPC (Virtual Private Cloud), Security Group & NACL (Network Access Control List), IAM (Identity and Access Management), EBS (Elastic Block Storage) and Ephemeral Storage, EFS (Elastic File System), S3 (Simple Storage Service) and S3 Glacier, EC2 (Elastic Compute Cloud), Database: RDS instance, DynamoDB	PPT + LABS		How do you design and secure a cloud infrastructure on AWS? Explain the role of a Virtual Private Cloud (VPC) , Security Groups and NACLs in managing network security, and describe how IAM is used to control access. Additionally, discuss storage options including EBS (and ephemeral storage), EFs , and S3/S3 Glacier ; outline the functions of EC2 for compute resources; and compare database services such as RDS and DynamoDB .
23			Amazon EC2 Part 2-Hindi/Urdu 	ELB (Elastic Load Balancing), Route 53 (DNS), Auto Scaling, CloudFront, CloudWatch, SNS, WorkMail, CloudFormation, Lambda, ECS (Elastic Container Service), ECR (Elastic Container Registry), EKS (Elastic Kubernetes Service)	PPT + LABS		Explain the key AWS services for improving application scalability, availability, and management. Discuss the role of Elastic Load Balancing (ELB) , Route 53 (DNS) , and Auto Scaling in maintaining high availability. How do CloudFront , CloudWatch , SNS , and WorkMail enhance performance monitoring, content delivery, and communication? Additionally, describe how CloudFormation automates infrastructure provisioning, and how Lambda , ECS , ECR , and EKS enable serverless computing and container management in AWS.
24			1. Terraform (In Hindi) - Introduc	Introduction to IaC - Infrastructure as Code, Introduction to Terraform, Working with Terraform CLI, Exploring Terraform commands	PPT + LABS		What is Infrastructure as Code (IaC) , and how does it benefit cloud infrastructure management? Introduce Terraform as a tool for IaC, and explain how to use the Terraform CLI for automating infrastructure tasks. Additionally, discuss key Terraform commands and their functions in managing infrastructure resources.
25			2. Terraform (In Hindi) - WhatW	Terraform modules, Terraform workflow, Planning & Writing in Terraform	PPT + LABS		What are Terraform modules , and how do they help in organizing and reusing infrastructure code? Explain the Terraform workflow , including the stages of initialization, planning, and application. Additionally, discuss the process of planning and writing in Terraform, and how it ensures efficient and predictable infrastructure management.

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26	Module 7	Terraform	3. Terraform (In Hindi) - How to	Terraform Create & Destroy, Terraform state on local backend, Terraform state on S3 buckets, Terraform create VPC	PPT + LABS	Learn Infrastructure as Code (IaC) and automate cloud infrastructure provisioning using Terraform.	How does Terraform handle infrastructure create and destroy operations, and what are the best practices for managing state? Explain the differences between storing Terraform state on a local backend and an S3 bucket , and how each method affects collaboration and state management. Additionally, describe how to use Terraform to create a VPC and the steps involved in configuring the network infrastructure.
27			4. Terraform (In Hindi) - How to	Terraform authentication methods, Variables, Available configurations in Terraform, Functions in Terraform	PPT + LABS		What are the different authentication methods available in Terraform for accessing cloud providers, and how do you configure them? Explain the use of variables in Terraform, including how they enhance flexibility in managing infrastructure. Additionally, discuss the available configurations and functions in Terraform, and how they simplify the process of writing reusable and efficient infrastructure code.
28	Module 8	GIT/GIT HUB ,GIT LAB	Git and GitHub Tutorial for Beginners	Git Workflow, Working Locally with Git, Working Remotely with GitHub	PPT + LABS	Gain expertise in version control with Git and collaborative project management using GitHub and GitLab.	What is the Git workflow , and how do you manage version control while working locally with Git? Explain the process of collaborating remotely using GitHub , including how to push, pull, and synchronize changes between local and remote repositories.
29				Deploy the Files to GitHub via Git, Branching and Merging	PPT + LABS		How do you deploy files to GitHub using Git, and what is the process for branching and merging in Git? Explain how to create and manage branches, resolve merge conflicts, and integrate changes from different branches into the main repository.
30				Resolve Merge Conflict, Git Reset and Stash Operation	PPT + LABS		How do you resolve merge conflicts in Git, and what are the differences between using git reset and git stash ? Explain scenarios where each command is useful in managing changes during version control.
31				How to Setup Git on Premises Hardware	PPT + LABS		How do you set up Git on on-premises hardware , and what are the key steps and configurations required to host and manage a private Git repository within a local network?
32	Module 9	JENKINS CI/CD	Jenkins CI & CD Introduction Part - 1 On	Introduction to Jenkins, Install and Setup Jenkins, Plugin Management in Jenkins	PPT + LABS	Understand Continuous Integration and Continuous Deployment (CI/CD) concepts and automation using Jenkins.	What is Jenkins , and how do you install and set it up for continuous integration? Additionally, explain the importance of plugin management in Jenkins and how plugins enhance its functionality.
33			Jenkins Installation Part - 2 On	Project Test and Auto deployment on Application Server, Build Pipeline View Project	PPT + LABS		How does Jenkins handle project testing and automated deployment to an application server, and how can you visualize and manage the process using the Build Pipeline View ?
34			Jenkins First Job Creation For Java	Configure Remote tasks Using Jenkins, Jenkins Slave Node Configuration	PPT + LABS		How do you configure remote tasks in Jenkins, and what is the process of setting up and managing Jenkins slave nodes to distribute workloads across multiple machines?
35			Jenkins Job Creation with Maven	Generate Reports & Enable Mail Notification, Integrations with Ansible, Docker & GIT Server	PPT + LABS		How can you generate reports and enable mail notifications in Jenkins for build status updates? Additionally, explain how Jenkins can be integrated with Ansible, Docker, and a Git server to automate deployment and infrastructure tasks.

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36	Module 10	MAVAEN	What Is Maven? What Is Maven	Core concepts of Maven	PPT + LABS	Learn to automate project build, dependency management, and deployment for Java-based applications using Maven.	What are the core concepts of Maven, and how does it help in managing project builds, dependencies, and the overall software development lifecycle? Let me know if you'd like to dive into any specific concept like the POM file or Maven lifecycle!
37			Maven Tutorial For Beginners In	Maven build automation, Maven dependency management	PPT + LABS		How does Maven automate the build process , and what is the role of dependency management in ensuring project consistency and efficiency?
38			Maven Installation On Mac How	Maven deployment automation	PPT + LABS		How does Maven support deployment automation , and what are the steps involved in deploying artifacts to a remote repository or application server using Maven?
39			Maven And Jenkins Integration B	Detailed examples of building, packaging, and deploying web and enterprise application projects using Maven, Maven IDE integration	PPT + LABS		How do you use Maven to build, package, and deploy web and enterprise applications , and how does IDE integration with tools like Eclipse or IntelliJ streamline these processes?
Project Module (1 Aug. to 30 Nov.)							
Project Task			Tentative Dates				
Project Assignment			20 July-31 July				
Progress-1							
Progress-2							
Progress-3							